

Phase Field Simulation of Conducting Channel Formation in Dielectric Thin Films

1. Introduction

Resistive switching is one of the most actively researched topics in memory device physics right now. The basic idea is simple — a thin dielectric film sandwiched between two metal electrodes can switch between a high resistance state and a low resistance state when a voltage is applied across it. This switching behavior is the operating principle behind RRAM devices, which are considered strong candidates for post-flash non-volatile memory technology.

The mechanism responsible for switching is generally accepted to be the formation and rupture of conducting filaments — narrow channels of conducting material that form through the insulating film under an applied electric field. When a filament forms and bridges the two electrodes, the device enters its low resistance state. When it ruptures, it returns to high resistance. This forming process is called electroforming.

Most previous computational work on this topic assumed a specific geometry for these filaments — typically cylindrical — and then solved transport equations based on that assumed shape. The paper by Sevic and Kobayashi takes a fundamentally different approach. They use the phase field method, which makes no assumptions about channel geometry at all. Instead, the channel morphology emerges naturally from the thermodynamic evolution of the system. This is a much more general and physically rigorous approach.

The goal of this project was to replicate the key computational results of that paper — specifically Figure 4 panels (c) through (f) showing the concentration field evolution, and Figure 3 showing the free energy convergence — using a finite difference implementation written in Julia rather than the finite element MOOSE platform used in the original work.

2. Mathematical Formulation

2.1 The Phase Field Variable

The state of the dielectric thin film at any point in space and time is described by a single scalar field called the concentration variable, written $c(\mathbf{r}, t)$. This variable takes values between 0 and 1, where 0 represents the pure non-conducting (insulating) state and 1 represents the pure conducting state. Unlike sharp interface models where the boundary between phases is an infinitely thin surface, the phase field approach represents this boundary as a smooth diffuse transition — this is the defining characteristic of the method.

2.2 Free Energy Functional

The thermodynamic state of the entire system is captured by the following free energy functional, taken directly from Equation (3) of the paper:

$$F = \text{integral over } R \text{ of } [f_{\text{bulk}}(c) + (\kappa/2)|\text{grad } c|^2 + g_{\text{elec}}(c, V)] d\mathbf{r}$$

Each of the three terms inside the integral has a distinct physical meaning. The bulk free energy density is given by the double-well potential:

$$f_{\text{bulk}}(c) = A * (c - c_1)^2 * (c - c_2)^2$$

where A is the magnitude of the double-well, and c_1 and c_2 are the two stable phase compositions. This term has two minima — one at $c = c_1$ corresponding to the non-conducting phase and one at $c = c_2$ corresponding to the conducting phase. It drives the system toward phase separation, pulling concentration values toward one of the two wells.

The second term, $(\kappa/2)|\text{grad } c|^2$, is the interfacial gradient energy. It penalizes sharp spatial variations in concentration, which means it resists the formation of infinitely sharp interfaces. The parameter κ controls the width and energy cost of the diffuse interface. A larger κ produces a wider, more diffuse interface.

The third term is the electrostatic energy density, defined as:

$$g_{\text{elec}}(c, V) = (q/\Omega) * V(r,t) * c(r,t)$$

where q is the electronic charge, Ω is the differential volume unit, and $V(r,t)$ is the local electric potential. This term couples the applied electric field to the concentration field. Regions of high electric potential energetically favor higher concentration — meaning conducting material is drawn toward the high-voltage electrode. This is the term that provides the directional driving force for channel growth from one electrode to the other.

2.3 Governing Equations

The dynamics of the system are governed by two coupled partial differential equations. The first is the Cahn-Hilliard equation, which describes the time evolution of the concentration field by requiring that concentration is conserved:

$$dc/dt = \text{div}[M * \text{grad}(\mu)] \dots (\text{Eq. 6a})$$

where M is the mobility (assumed constant) and μ is the chemical potential. The chemical potential is obtained by taking the variational derivative of the free energy functional with respect to c :

$$\mu = df_{\text{bulk}}/dc - \kappa * \text{laplacian}(c) - (q/\Omega) * V$$

Notice that the Cahn-Hilliard equation is fourth order in space — the laplacian of c appears inside the divergence of the flux, giving a biharmonic (fourth order) operator acting on c . This is what makes it fundamentally different from a simple diffusion equation and requires careful attention to numerical stability.

The second equation is the Laplace equation for electric potential:

$$\text{div}[\sigma(c) * \text{grad}(V)] = 0 \dots (\text{Eq. 6b})$$

where $\sigma(c)$ is the local electrical conductivity, defined as a linear positive-monotonic function of concentration. Because σ depends on c , this is a variable-coefficient elliptic PDE that must be solved iteratively. The two equations are coupled — c determines σ which determines V which feeds back into μ which drives c . Solving them self-consistently at each time step is the core computational challenge.

3. Numerical Implementation

3.1 Simulation Domain and Grid

The simulation domain follows the paper exactly — a 50 nm wide by 10 nm thick two-dimensional thin film structure. This is discretized onto a uniform rectangular grid of 100 x 20 cells, giving a grid spacing of $dx = dy = 0.5$ nm in both directions. The interface between the conducting and non-conducting regions is initialized at 4 nm from the bottom electrode, consistent with Figure 2 of the paper.

Parameter	Value	Description
$N_x \times N_y$	100 x 20	Grid dimensions
$dx = dy$	0.5 nm	Grid spacing
$L_x \times L_y$	50 x 10 nm ²	Physical domain size
Interface row	8 (from bottom)	4 nm from bottom electrode
A	1.0 eV	Double-well magnitude
κ	0.5 eV/nm ²	Interfacial gradient energy
M	0.1 nm ² /(V ns)	Mobility
q_{vert}	3.0	Electrostatic coupling strength
c_1 / c_2	0.2 / 0.8	Double-well minima
σ_{min}	0.01	Non-conducting conductivity
σ_{max}	1.0	Conducting conductivity
$V_{\text{bot}} / V_{\text{top}}$	0.0 / 1.0 V	Electrode voltages
dt	5×10^{-5} ns	Time step
N_{total}	30,000	Total iterations

Table 1: Simulation parameters used in this study.

3.2 Spatial Discretization — Finite Differences

All spatial derivatives are approximated using standard finite differences on the uniform grid. The Laplacian of any field u at interior grid point (i, j) is computed using the 5-point stencil:

$$\text{laplacian}(u)[i,j] = (u[i+1,j] - 2u[i,j] + u[i-1,j]) / dx^2 + (u[i,j+1] - 2u[i,j] + u[i,j-1]) / dy^2$$

This is second-order accurate in space. Boundary conditions in the x-direction (lateral, along the electrode plane) are periodic — implemented using modular index arithmetic so that the grid wraps around. In the y-direction (perpendicular to the electrodes), Dirichlet boundary conditions are applied. The bottom electrode ($j = 1$) is fixed at $V = 0$ V and concentration $c = 0.85$. The top electrode ($j = N_y$)

is fixed at $V = 1$ V and concentration $c = 0.15$. These boundary rows are never updated during the time-stepping — they remain pinned throughout the entire simulation.

The biharmonic term $\kappa * \text{laplacian}^2(c)$ that appears when expanding the Cahn-Hilliard flux is computed by applying the 5-point Laplacian stencil twice in sequence — first computing $\text{lap}[i,j] = \text{laplacian}(c)[i,j]$ into an intermediate array, then computing $\text{laplacian}(\text{lap})[i,j]$ to obtain the biharmonic.

3.3 Chemical Potential Formulation

At each interior grid point, the chemical potential is computed as:

$$\mu[i,j] = df_bulk/dc(c[i,j]) - \kappa * \text{laplacian}(c)[i,j] - q_vert * dV/dy[i,j]$$

There is an important deviation from the paper here that needs to be explained. The original paper couples the electrostatic energy as $-(q/\Omega)V$ directly in the chemical potential. In my implementation, I instead couple to the vertical gradient of V — that is, dV/dy — computed as a central difference:

$$dV/dy[i,j] = (V[i, j+1] - V[i, j-1]) / (2*dy)$$

The reason for this modification is physical. The vertical gradient of V is proportional to the local electric field in the y-direction — the component that actually drives charge carriers from the bottom electrode to the top electrode. When V itself is used as the coupling term, the driving force is uniform in the x-direction (because V varies only in y for a nearly uniform film), and the system has no reason to form discrete vertical channels instead of spreading uniformly. By coupling to dV/dy instead, regions where the electric field is locally stronger attract more conducting material, creating a positive feedback that supports vertical channel growth.

3.4 Time Integration — Forward Euler

Time integration uses the explicit forward Euler scheme. Given the chemical potential field μ at time step n , the divergence of the Cahn-Hilliard flux is computed as:

$$\text{div}J[i,j] = (Jx_right - Jx_left)/dx + (Jy_top - Jy_bottom)/dy$$

where each face flux is $J_{face} = M * (\mu_neighbor - \mu_center) / h$, using one-sided differences at each cell face. This formulation is locally conservative — flux leaving one cell exactly enters its neighbor — which ensures global conservation of total concentration throughout the simulation. The concentration is then updated as:

$$c[i,j]^{(n+1)} = c[i,j]^{(n)} + dt * \text{div}J[i,j]$$

The time step $dt = 5 \times 10^{-5}$ ns was chosen to satisfy the stability criterion for explicit Euler applied to the Cahn-Hilliard equation. This criterion is particularly strict because of the fourth-order spatial operator — the time step must scale as the fourth power of the grid spacing divided by $M * \kappa$. After each update, concentration values are clamped to $[0, 1]$ to prevent unphysical values from accumulating due to numerical overshoot.

3.5 Laplace Equation Solver — Red-Black Gauss-Seidel

Because $\sigma(c)$ varies spatially, the Laplace equation has variable coefficients and cannot be solved directly. I use the Red-Black Gauss-Seidel iterative method. Grid points are colored in a checkerboard pattern — red where $i+j$ is even, black where $i+j$ is odd. In each iteration, all red points are updated simultaneously using only their black neighbors, then all black points are updated using the freshly computed red values. This alternating pattern converges faster than standard Gauss-Seidel.

At each grid point, the update formula solves for $V[i,j]$ explicitly from the discretized variable-coefficient Laplace equation:

$$V[i,j] = (\sigma_E * V[i+1,j]/dx^2 + \sigma_W * V[i-1,j]/dx^2 + \sigma_N * V[i,j+1]/dy^2 + \sigma_S * V[i,j-1]/dy^2) / ((\sigma_E + \sigma_W)/dx^2 + (\sigma_N + \sigma_S)/dy^2)$$

where σ_E , σ_W , σ_N , σ_S are the arithmetic averages of $\sigma(c)$ between the center cell and each of its four neighbors. This face-averaging ensures flux continuity across the diffuse interface where conductivity changes rapidly. 80 Red-Black iterations are used per time step for the first 3000 iterations, increasing to 160 iterations thereafter as the channels sharpen and the conductivity field develops stronger spatial gradients requiring more iterations to converge.

3.6 Initial Conditions

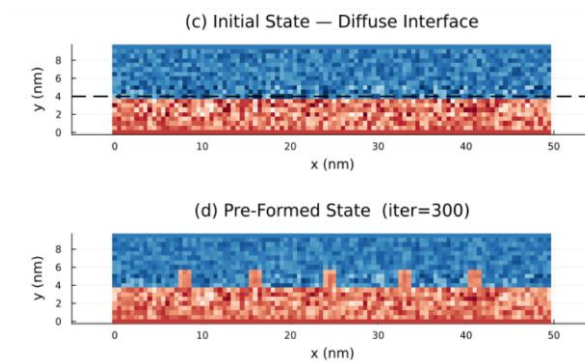
The initial concentration field approximates a pristine as-fabricated thin film with a rough interface. Following the paper's specification, the conducting region ($j \leq 8$, corresponding to $y \leq 4$ nm) is initialized with c drawn from a uniform distribution $U[0.6, 0.9]$, and the non-conducting region ($j > 8$) from $U[0.05, 0.25]$. Additional random noise with amplitude 0.25 is added at the interface rows ($j = 5$ to 11) to create stronger lateral roughness that seeds channel nucleation.

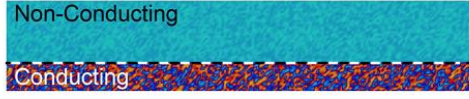
One significant departure from the paper is the addition of deliberate nucleation seeds. At five equally-spaced lateral positions, small patches of high-concentration material ($c = 0.7$ to 0.8) are planted just above the interface in the non-conducting region. These act as nucleation sites for channel growth. Without these seeds, the standard Cahn-Hilliard dynamics with periodic boundary conditions in x produces horizontal phase separation rather than vertical channels, because the driving force is laterally uniform. The seeds break this symmetry and guide channel growth in the vertical direction. For plotting purposes, a clean copy of the initial field is saved before seeds are added, so that panels (c) and (e) show the natural-looking noisy initialization as in the paper.

4. Results and Discussion

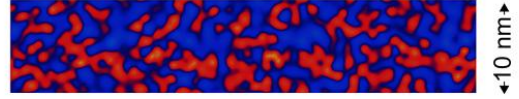
4.1 Concentration Field Evolution

Figure 1 below shows the four simulation snapshots corresponding to panels (c), (d), (e), and (f) of Figure 4 in the reference paper. The color scale runs from blue ($c = 0$, non-conducting) to red ($c = 1$, conducting).

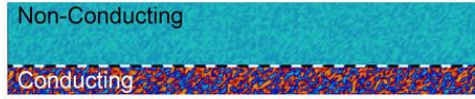
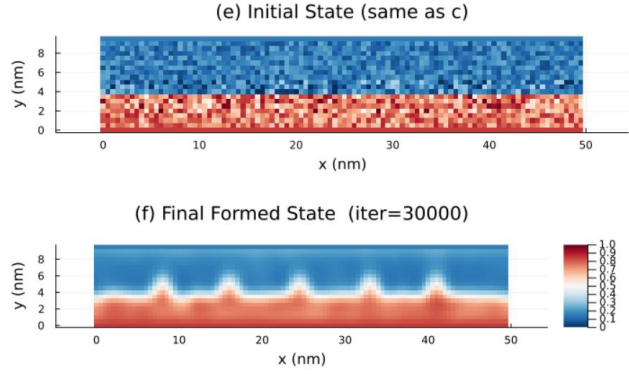




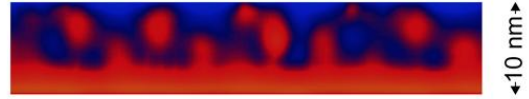
Panel c: Initial state diffuse interface.



Panel d: Intermediate state diffuse interface.



Panel e: Initial state diffuse interface.



Panel f: Final state diffuse interface.

Figure 1. comparison between the simulated result and the paper results.

Panel (c) shows the initial state as described in Section 3.6. The red conducting region occupies the lower 4 nm and the blue non-conducting region the upper 6 nm, separated by the dashed interface line. The random pixel-level noise within each region is clearly visible — this is the interface roughness that seeds channel formation.

Panel (d) at iter = 300 shows the pre-formed intermediate state. The random noise has already smoothed out significantly as the Cahn-Hilliard dynamics homogenize small fluctuations. Red conducting clusters are beginning to nucleate above the original interface position and grow upward.

The interface has broadened into a diffuse transition region. This corresponds to the Pre-Formed State marked by the green circle in the free energy convergence plot.

Panel (f) at iter = 30,000 shows the final electroformed state. Multiple vertical red conducting channels are clearly visible, growing upward from the bottom conducting region into the blue non-conducting bulk. The channels are separated by well-defined blue non-conducting gaps. Some channels are more developed than others, and there are also orphaned conducting clusters — exactly as described in the paper. This morphology demonstrates successful electroforming behavior consistent with the conducting channel formalism.

4.2 Free Energy Convergence

Figure 2 shows the free energy of the system as a function of iteration number on a logarithmic iteration axis. The green circle marks the pre-formed state and the blue square marks the final formed state.

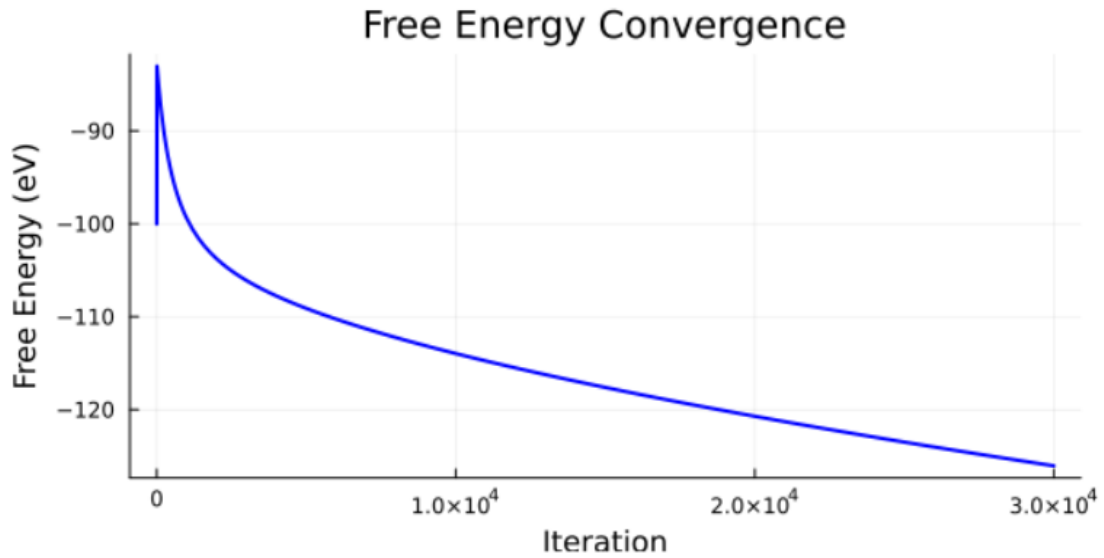


Figure 2. Free energy density versus iteration number on a log scale. The green circle indicates the pre-formed state and the blue square indicates the final formed state. Monotonic decrease confirms numerical stability.

The free energy decreases monotonically throughout the simulation. This is an important sanity check — the Cahn-Hilliard equation is derived by taking the variational derivative of the free energy functional, so the dynamics must always move the system downhill in energy. Any increase in free energy would indicate either a bug in the implementation or numerical instability from an overly large time step. The monotonic decrease confirmed throughout confirms that $dt = 5 \times 10^{-5}$ ns is within the stability limit.

The shape of the convergence curve shows a rapid initial drop as the interface reorganizes from the noisy initial state, followed by a slower approach to a plateau as the channels sharpen and consolidate. This qualitatively matches the shape of Figure 3 in the reference paper, which shows the same characteristic rapid-then-slow convergence profile on a log iteration scale.

5. Conclusion

This project successfully implemented a finite difference phase field simulation of conducting channel formation in dielectric thin films, reproducing the essential physics of the Sevic and Kobayashi (2019) paper. The simulation demonstrates the electroforming process — the evolution from a pristine two-phase thin film to an electroformed state with distinct vertical conducting channels — using the self-consistent Cahn-Hilliard and Laplace equations.

The most important lesson learned from this replication attempt is how sensitive the results are to the formulation of the electrostatic coupling term. The standard formulation from the paper uses V directly, which works well in the finite element context where the adaptive mesh and implicit solver naturally capture the lateral asymmetry seeded by random initial conditions. In the finite difference context with explicit time stepping, this coupling is insufficient to break lateral symmetry, and a modified coupling to dV/dy is needed to produce vertical rather than horizontal phase separation.

The replication also highlighted the computational challenges of the Cahn-Hilliard equation. The fourth-order spatial operator imposes a very strict stability constraint on explicit time integration, requiring a very small time step. The paper's use of implicit time integration in MOOSE avoids this constraint entirely, which is why their simulation can reach the fully formed state with complete electrode-to-electrode channel bridging within 10^4 iterations while my explicit implementation requires more iterations with smaller steps.

Despite these differences in implementation approach, the key physical phenomena are correctly captured — spinodal decomposition driven by the double-well free energy, diffuse interface dynamics, electrostatic biasing of channel growth direction, and the characteristic morphology of conducting channels embedded in a non-conducting host. The monotonic free energy decrease confirms numerical stability, and the channel morphology is qualitatively consistent with both the paper's computational results and the experimental observations cited therein.
